KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define and discuss goals of computer security. Using a block diagram show relation and explain terminologies of any cryptosystem. (08) [CO1]
   b) Discuss Shannon characteristics to describe characteristics of good cipher. "Ciphertext size should not be larger than plaintext" – show how RSA and ElGamal cryptosystems retain this criterion. (14) [CO2]
   c) Define and discuss characteristics of zero-knowledge proof. How and why digital signatures with public key cryptography is required to combine? Explain the procedure. (13) [CO2]

2. a) Using block diagrams show that asymmetric cryptosystem is alluring than symmetric cryptosystem. Illustrate the major steps of each cycle of AES algorithm using an appropriate diagram. (13) [CO1]
   b) Discuss characteristics of cryptographic hash function. Asymmetric cryptosystem and hash technique need to combine concurrently – why and how? (13) [CO1]
   c) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (09) [CO2]

3. a) Discuss properties retained by digital signatures (DS). Derive the steps of ElGamal digital signature. Show that it is capable to satisfy properties of DS. (13) [CO3]
   b) Define the term cryptanalysis. Explain three terms allied with attack on cryptanalysis. (10) [CO1]
   c) Explain the working procedure of Double DES and Triple DES. Establish a comparative analysis between AES and DES based on various features of design. (12) [CO2]

4. a) ElGamal cipher can be developed to form 'product cipher' and 're-randomized cipher'. How? Show step by step. (14) [CO4]
   b) Derive encryption-decryption procedure of Caesar cipher and Vernam cipher. For RSA algorithm from $ed \equiv 1 \bmod \varphi(n)$, establish that many values may exist for d for a single value of e, where symbols have their usual meanings. (14) [CO3]
   c) Derive a bonding among order, prime and primitive element considering the prime 13 and order 2. (07) [CO4]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are covert channels? Differentiate between error and fault. (08) [CO2]
   b) What is 'Salt' in rainbow table attack? How does viruses gain control in file directory? (10) [CO1]
   c) What is segmentation fault in buffer overflow attack? Explain with figure. Also, discuss the countermeasures of buffer overflow attack. (10) [CO2]
   d) How webcams are used in Distributed Denial of Service (DDOS) attack? (07) [CO3]

6. a) Explain how paging removes the drawbacks of segmentation. (08) [CO3]
   b) Show how integer overflow attack compromises a program's reliability and security. (15) [CO4]
   c) How does a Demilitarized Zone (DMZ) network work? "Paillier cryptosystem permits addition of two ciphertexts, or one ciphertext and one plaintext or multiplication of one ciphertext and one plaintext" – justify it. (12) [CO4]

7. a) "Format string vulnerability is responsible to crash a program" – explain it. (05) [CO2]
   b) Explain relocation with example. What are the advantages of segmentation in operating system? (10) [CO2]
   c) Write short notes on the followings: (12) [CO1]
   (i) Trojan horse, (ii) Boot sector viruses, (iii) Dictionary attack.
   d) What is macro? Discuss how does document virus attack work in excel file using Dynamic Data Exchange (DDE). (08) [CO4]

CamScanner

8. a) What is a virus signature? How does antivirus work on a system? (10) [CO2]
   b) Differentiate between Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). (10) [CO2]
   c) What is light weight cryptography? Give an example and working procedure of it. (10) [CO4]
   d) Describe Stateful Inspection firewalls. (05) [CO1]

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